

Language Python 3  

main.py

```
1 def top_k_scores(arr, k):
2
3     arr = arr.copy()
4
5     for i in range(min(k, len(arr))):
6
7         max_index = i
8
9         for j in range(i + 1, len(arr)):
10             if arr[j] > arr[max_index]:
11                 max_index = j
12
13         arr[i], arr[max_index] = arr[max_index], arr[i]
14
15     return arr[:k]
16
17
18 scores = [72,88,65,90,77,95,60,83,91,68]
19
20 print("Top 5 Scores:", top_k_scores(scores,5))
```



input

Top 5 Scores: [95, 91, 90, 88, 83]

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Press ENTER to exit console.

main.py

```

1- def insertion_sort_count_shifts(arr):
2     shifts = 0
3     for i in range(1, len(arr)):
4         key = arr[i]
5         j = i - 1
6         while j >= 0 and arr[j] > key:
7             arr[j + 1] = arr[j]
8             shifts += 1
9             j -= 1
10        arr[j + 1] = key
11    return arr, shifts
12 log = [18.2, 18.5, 18.9, 17.9, 19.1, 19.4, 19.0]
13 sorted_log, shifts = insertion_sort_count_shifts(log)
14 print("Sorted Log:", sorted_log)
15
16 print("Total Shifts:", shifts)

```

Sorted Log: [17.9, 18.2, 18.5, 18.9, 19.0, 19.1, 19.4]
Total Shifts: 5

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```
main.py
1 def insert_song(playlist, song):
2     playlist.append(song)
3     i = len(playlist) - 2
4     while i >= 0 and playlist[i][1] > song[1]:
5         playlist[i + 1] = playlist[i]
6         i -= 1
7     playlist[i + 1] = song
8     return playlist
9 playlist = [("Intro",120),
10            ("Chill Beat",210),
11            ("Long Jam",340)]
12 new_song = ("Quick Track",180)
13 updated = insert_song(playlist, new_song)
14 print(updated)
```

input

```
('Intro', 120), ('Quick Track', 180), ('Chill Beat', 210), ('Long Jam', 340)]

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```

main.py

```
1- def insert_price(prices, price):
2     prices.append(price)
3     i = len(prices) - 2
4-     while i >= 0 and prices[i] > price:
5         prices[i + 1] = prices[i]
6         i -= 1
7     prices[i + 1] = price
8     return prices
9 prices = []
10 data = [102.5, 98.3, 105.1, 100.0, 97.8]
11- for p in data:
12     prices = insert_price(prices, p)
13 print("Sorted Prices:", prices)
```

Sorted Prices: [97.8, 98.3, 100.0, 102.5, 105.1]

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Run

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main.py

1- def insert_updated_score(board, score):

2 shifts = 0

3 board.append(score)

4 i = len(board) - 2

5- while i >= 0 and board[i] < score:

6 board[i + 1] = board[i]

7 shifts += 1

8 i -= 1

9 board[i + 1] = score

10 return board, shifts

11 board = [980, 875, 760, 690, 500]

12 result, shifts = insert_updated_score(board, 820)

13 print("Updated Leaderboard:", result)

14 print("Shifts:", shifts)

input

Updated Leaderboard: [980, 875, 820, 760, 690, 500]

Shifts: 3

...

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main.py

```
1 def bubble_sort_hand(hand):
2     passes = 0
3     for i in range(len(hand)-1):
4         swapped = False
5         passes += 1
6         for j in range(len(hand)-1-i):
7             if hand[j] > hand[j+1]:
8                 hand[j], hand[j+1] = hand[j+1], hand[j]
9                 swapped = True
10
11         if swapped == False:
12             break
13     return hand, passes
14 hand = [2,4,6,8,9,11,13]
15 hand.append(7)
16 sorted_hand, p = bubble_sort_hand(hand)
17
18 print("Sorted Hand:", sorted_hand)
19 print("Passes:", p)
```

Sorted Hand: [2, 4, 6, 7, 8, 9, 11, 13]
Passes: 5

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Run

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main.py

```
1 def bubble_sort(arr):
2     comparisons = 0
3     for i in range(len(arr)-1):
4         swapped = False
5         for j in range(len(arr)-1-i):
6             comparisons += 1
7             if arr[j] > arr[j+1]:
8                 arr[j], arr[j+1] = arr[j+1], arr[j]
9                 swapped = True
10
11         if swapped == False:
12             break
13     return arr, comparisons
14 alerts = [2,1,3,2,1,4,3,2,5,1,2,3,4,1,2]
15
16 result, count = bubble_sort(alerts)
17
18 print("Sorted Alerts:", result)
19 print("Comparisons:", count)
```

Sorted Alerts: [1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 4, 4, 5]
Comparisons: 99

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main.py

```
1 def bubble_sort_with_frames(arr):
2
3     print("Original:", arr)
4
5     for i in range(len(arr)-1):
6
7         for j in range(len(arr)-1-i):
8
9             if arr[j] > arr[j+1]:
10                 arr[j], arr[j+1] = arr[j+1], arr[j]
11
12         print("Pass", i+1, ":", arr)
13
14     return arr
15
16
17 numbers = [5,1,4,2,8]
18
19 bubble_sort_with_frames(numbers)
```

input

```
Original: [5, 1, 4, 2, 8]
Pass 1 : [1, 4, 2, 5, 8]
Pass 2 : [1, 2, 4, 5, 8]
Pass 3 : [1, 2, 4, 5, 8]
Pass 4 : [1, 2, 4, 5, 8]
```

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```

main.py

```
1 def bubble_sort_queue(queue):
2
3     priority = {
4         "ambulance":1,
5         "bus":2,
6         "car":3
7     }
8
9     for i in range(len(queue)-1):
10         for j in range(len(queue)-1-i):
11
12             if priority[queue[j]] > priority[queue[j+1]]:
13                 queue[j], queue[j+1] = queue[j+1], queue[j]
14
15     return queue
16
17
18 queue = ["car","car","bus"]
19
20 queue.append("ambulance")
21
22 print(bubble_sort_queue(queue))
```

```
['ambulance', 'bus', 'car', 'car']
```

```
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```

```

main.py
1 def optimized_bubble_sort(arr):
2
3     passes = 0
4
5     for i in range(len(arr)-1):
6
7         swapped = False
8         passes += 1
9
10        for j in range(len(arr)-1-i):
11            if arr[j] > arr[j+1]:
12                arr[j], arr[j+1] = arr[j+1], arr[j]
13                swapped = True
14
15        if swapped == False:
16            break
17
18    return arr, passes
19
20
21 rolls = [101,102,104,103,105,107,106,108]
22
23 result, p = optimized_bubble_sort(rolls)
24
25 print("Sorted Roll Numbers:", result)
26 print("Passes:", p)
    
```

input

```

Sorted Roll Numbers: [101, 102, 103, 104, 105, 106, 107, 108]
Passes: 2
    
```

```

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```

```
main.py
1 import time
2 def selection_sort(arr):
3     for i in range(len(arr)):
4         min_index = i
5         for j in range(i+1, len(arr)):
6             if arr[j] < arr[min_index]:
7                 min_index = j
8         arr[i], arr[min_index] = arr[min_index], arr[i]
9     return arr
10 prices = [499,129,899,45,275,60,310,150]
11 start = time.time()
12 result = selection_sort(prices)
13 end = time.time()
14 print("Sorted Prices:", result)
15 print("Time Taken:", end - start, "seconds")
```

Sorted Prices: [45, 60, 129, 150, 275, 310, 499, 899]
Time Taken: 1.2636184692382812e-05 seconds

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```
main.py
1 def distribute_prizes(players):
2     for i in range(len(players)):
3         max_index = i
4         for j in range(i+1, len(players)):
5             if players[j][1] > players[max_index][1]:
6                 max_index = j
7         players[i], players[max_index] = players[max_index], players[i]
8     return players
9 players = [("Asha",88),("Ravi",95),("Meera",79),("Dev",95)]
10
11 ranking = distribute_prizes(players)
12
13 print("Prize Ranking")
14
15 for name, score in ranking:
16     print(name, score)
```

input

Prize Ranking

Ravi 95

Dev 95

Asha 88

Meera 79

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main.py

```
1 def reorder_shelf(arr):
2     moves = 0
3
4     for i in range(len(arr)):
5         min_index = i
6         for j in range(i+1, len(arr)):
7             if arr[j] < arr[min_index]:
8                 min_index = j
9
10        if min_index != i:
11            arr[i], arr[min_index] = arr[min_index], arr[i]
12            moves += 1
13    return arr, moves
14 books = [305,102,250,118,199,400,101]
15 sorted_books, moves = reorder_shelf(books)
16 print("Sorted Books:", sorted_books)
17 print("Physical Moves:", moves)
```

Sorted Books: [101, 102, 118, 199, 250, 305, 400]

Physical Moves: 4

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Language Python 3  

main.py

```
1 def selection_sort_min_writes(arr):
2
3     swaps = 0
4
5     for i in range(len(arr)):
6
7         min_index = i
8
9         for j in range(i+1, len(arr)):
10             if arr[j] < arr[min_index]:
11                 min_index = j
12
13             if min_index != i:
14                 arr[i], arr[min_index] = arr[min_index], arr[i]
15                 swaps += 1
16
17     return arr, swaps
18
19
20 data = [23.5,19.2,25.1,18.8,21.4]
21
22 sorted_data, swap_count = selection_sort_min_writes(data)
23
24 print("Sorted List:", sorted_data)
25 print("Total Swaps:", swap_count)
```



input

```
Sorted List: [18.8, 19.2, 21.4, 23.5, 25.1]
Total Swaps: 2
```

```
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```